Windows Domain Login + Active Directory Access

Interview Preparation Notes — DHCP, DNS, Kerberos & LDAP

Scenario: Windows client → company.com domain → DC01.company.com → user Shambhu

Core flow:

```
DHCP → DNS → Kerberos AS → TGT → Kerberos TGS → LDAP Service Ticket → LDAP → Active Directory
Directory Information
```

1. DHCP — Getting Network Connectivity

When the Windows client needs dynamic network configuration, it contacts a DHCP server. The normal exchange is DHCP Discover → DHCP Offer → DHCP Request → DHCP ACK.

Configuration	Example
IP Address	192.168.1.50
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.10

Interview point: DHCP provides network configuration. It is not part of Kerberos authentication itself.

2. DNS — Finding the Domain Controller

The Windows client uses DNS to locate Active Directory services and Domain Controllers. Active Directory relies heavily on DNS and registers service (SRV) records for services such as LDAP and Kerberos.

```
_ldap._tcp.dc._msdcs.company.com
_kerberos._tcp.company.com
```

DNS answers the question: **“Where is the Domain Controller / AD service?”** DNS does not authenticate the user.

3. Kerberos — Initial User Authentication

The Domain Controller provides the Kerberos Key Distribution Center (KDC). Conceptually, the KDC contains two logical services: Authentication Service (AS) and Ticket Granting Service (TGS).

```
KDC
■■■ AS – Authentication Service
■■■ TGS – Ticket Granting Service
```

3A. Client → Authentication Service

The client begins the Kerberos Authentication Service exchange with an AS-REQ. The user's password is not transmitted across the network as plaintext.

```
Client → KDC / AS : AS-REQ
```

3B. KDC → Client: TGT

After successful authentication, the KDC returns an AS-REP containing a Ticket Granting Ticket (TGT) and related session information. The TGT allows the client to request service tickets without repeating the full initial authentication.

```
KDC / AS → Client : AS-REP + TGT
```

4. TGT Is Not Sent Directly to LDAP

This is the key distinction to remember in an interview. The client does **not** normally present the TGT directly to LDAP. The TGT is used with the Kerberos Ticket Granting Service to obtain a service ticket for the specific service the client wants to access.

```
TGT → TGS → LDAP Service Ticket → LDAP
```

5. TGS — Requesting an LDAP Service Ticket

When Windows needs to query Active Directory through LDAP, the client sends a TGS-REQ to the KDC/TGS, using its TGT to request a ticket for the LDAP service on the Domain Controller.

```
Client → KDC / TGS : TGS-REQ + TGT
KDC / TGS → Client : TGS-REP + LDAP Service Ticket
```

6. LDAP — Accessing Active Directory

LDAP (Lightweight Directory Access Protocol) is the directory access/communication protocol. After obtaining the LDAP service ticket, the client can authenticate to the LDAP service, commonly using SASL/GSSAPI with Kerberos, and then perform directory operations.

Example query: “Find Shambhu's group memberships.” The LDAP service processes the directory search and returns information such as users, groups, computers, and other directory attributes.

```
Client → LDAP service : authenticated LDAP request
LDAP service → Client : directory data / search response
```

7. Example: File Share Access

The same Kerberos idea applies to other services. If the user accesses a Windows file share such as \\FileServer\Finance, the client uses its TGT to obtain an SMB service ticket, then authenticates to the SMB service.

```
TGT → TGS → SMB Service Ticket → SMB → File Server
```

8. Complete End-to-End Flow

1. Windows client starts and needs network configuration.
2. DHCP provides IP address, subnet mask, gateway, and DNS server.
3. Client uses DNS to locate the Domain Controller and AD services through DNS/SRV records.
4. Client contacts the Kerberos Authentication Service and performs the AS exchange.
5. KDC authenticates the user and issues a TGT.
6. Client needs directory information and uses the TGT to request an LDAP service ticket from the TGS.
7. KDC returns the LDAP service ticket.
8. Client connects to LDAP and authenticates using the Kerberos service ticket, commonly via SASL/GSSAPI.
9. LDAP processes the directory query.
10. Active Directory returns the requested directory information.

```
DHCP
↓
IP + Gateway + DNS
↓
DNS
↓
Find Domain Controller
↓
Kerberos AS
↓
AS-REQ / AS-REP
↓
TGT
```

```
↓
Kerberos TGS
↓
TGS-REQ / TGS-REP
↓
LDAP Service Ticket
↓
LDAP
↓
Directory Queries
↓
Users / Groups / Computers / Attributes
```

9. One-Minute Interview Answer

“First, if the Windows client uses dynamic addressing, DHCP provides the IP address, subnet mask, default gateway and DNS server. The client then uses DNS to locate a Domain Controller by querying Active Directory DNS records, especially SRV records. Once the Domain Controller is located, Kerberos is used for authentication. The client performs the Authentication Service exchange and, after successful authentication, receives a Ticket Granting Ticket, or TGT. When the client needs to access a specific service such as LDAP, it uses the TGT to request a service ticket from the Kerberos Ticket Granting Service. The client then connects to LDAP and uses the LDAP service ticket for authentication, commonly through SASL/GSSAPI. LDAP handles the actual directory communication and queries, such as retrieving user, group, or computer information. So, DHCP provides network configuration, DNS locates the Domain Controller, Kerberos authenticates and provides tickets, and LDAP accesses the directory.”

10. Quick Memory Trick

Component	Main Question / Job
DHCP	“How do I get onto the network?”
DNS	“Where is my Domain Controller?”
Kerberos	“Who am I, and can I access this service?”
LDAP	“What information do I want from the directory?”

Most important correction: ■ TGT → LDAP directly ■ TGT → TGS → LDAP service ticket → LDAP

Final mental model: DHCP gives connectivity → DNS finds the DC → Kerberos proves identity and obtains tickets → LDAP retrieves directory information.